Mini-project 1 AM215

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Mathematical Model for Tennis

State Space Definition

A tennis game has a hierarchical scoring structure. Players score points to win games. They score games to win sets. They score sets to win matches.

For one game (the smallest unit after individual points), we define:

- Point Values: 0, 15, 30, 40 (we use indices 0, 1, 2, 3).
- States:
  - Regular states: (i, j) where $i, j \in \{0, 1, 2, 3\}$ represent (server points, receiver points).
  - Special states when both reach 40:
    - Deuce $\equiv (3, 3)$: tied at 40-40
    - Ad-S: advantage server (sometimes referred to as "ad in")
    - Ad-R: advantage receiver (sometimes referred to as "ad out")
  - Absorbing states:
    - Win-S: server wins game
    - Win-R: receiver wins game

Transition Rates

There is an important difference between a game like tennis and one like soccer or basketball. Rates depend much more significantly on who is serving, as opposed to score difference. This varies from player to player, but in general, especially at the professional level, players tend to win more games when they are serving than when they are receiving. This phenomenon gives rise to the terms "hold" and "break" in tennis. A player is said to *hold* when they successfully win a game while serving, reflecting the expectation that serving provides a substantial advantage. Conversely, a *break* occurs when the receiving player wins the game, which disrupts the server's advantage and often shifts the momentum of the match. For instance, the statistic "% service games won" (typically around 85% or higher for professional players) indicates how frequently a player holds serve. This metric can be explored on the ATP Tour Servers leaderboard.

- λ_s: rate at which server wins a point
- λ_r: rate at which receiver wins a point
- Total event rate: $\lambda = \lambda_s + \lambda_r$

As an illustrative example, suppose the server wins points at a rate of 0.7λ while the receiver wins at a rate of 0.3λ, so that the probability of the server winning any given point is

$$p = \frac{0.7\lambda}{0.7\lambda + 0.3\lambda} = 0.7, \qquad q = 1 - p = 0.3.$$

This 70% probability is based on the probability for the top player in the above ATP Tour serve leaderboard, which at the current moment is Giovanni Mpetshi Perricard. Note that he wins 79.2% of the points where his first serve goes in, and he makes his first serve in 65.1% of the time, whereas he only wins 53.9% of the points on his second serve. So, ignoring double faults for now (which tend to be very infrequent for top players), we get an approximate probability of him winning a serve point of: $0.792 * 0.651 + 0.539 * (1 - 0.651) \approx 0.70$, which is where the 70% figure comes from.

The probability that the server wins a service game (i.e. holds serve) can be calculated by considering both outcomes before and after deuce:

$$P_{\text{server win}} = P_{\text{pre}} + P_{\text{deuce}} \cdot P_{\text{deuce}\to\text{win}},$$

1. Winning before deuce. The server can win the game by reaching 4 points while the receiver has at most 2 points (if the receiver wins 3, it would be deuce). There are three scenarios:

- 4–0: occurs with probability p^4,
- 4–1: occurs with probability $\binom{4}{1} q p^4$,
- 4–2: occurs with probability $\binom{5}{2} q^2 p^4$.

Factoring p^4, the total probability is

$$P_{\text{pre}} = p^4 \left(1 + 4q + 10q^2\right).$$

For $p = 0.7$, $q = 0.3$, this evaluates to

$$P_{\text{pre}} = 0.7^4 \left(1 + 4(0.3) + 10(0.3)^2\right) \approx 0.744.$$

2. Reaching deuce. Deuce occurs when both players win exactly 3 points in the first 6 points played. The probability of this event is given by the binomial coefficient

$$P_{\text{deuce}} = \binom{6}{3} p^3 q^3.$$

With $p = 0.7$, $q = 0.3$, we obtain

$$P_{\text{deuce}} = 20 \cdot (0.7^3)(0.3^3) \approx 0.185.$$

3. Winning from deuce. Once at deuce, each sequence of two points can either:

- result in the server winning outright with probability p^2,
- result in the receiver winning outright with probability q^2,
- or return to deuce with probability $2pq$.

Let D be the probability the server wins from deuce.

$$D = p^2 + (2pq)D$$

$$D(1 - 2pq) = p^2$$

$$D = \frac{p^2}{1 - 2pq}$$

$$1 - 2pq = p^2 + q^2$$

$$D = \frac{p^2}{p^2 + q^2}$$

So the probability the server eventually wins from deuce is

$$P_{\text{deuce}\to\text{win}} = \frac{p^2}{p^2 + q^2}.$$

For $p = 0.7$, $q = 0.3$,

$$P_{\text{deuce}\to\text{win}} = \frac{0.7^2}{0.7^2 + 0.3^2} \approx 0.845.$$

4. Total probability of winning the service game. Combining both cases gives

$$P_{\text{server win}} = P_{\text{pre}} + P_{\text{deuce}} \cdot P_{\text{deuce}\to\text{win}}.$$

Substituting the numerical values,

$$P_{\text{server win}} \approx 0.744 + 0.185 \times 0.845 \approx 0.901.$$

So, with a point-winning probability of 70% on serve, the server is expected to hold serve about 90.1% of the time, which is similar to what we see for Giovanni Mpetshi Perricard.

Let's assume that for a typical strong server, $\lambda_s \approx 0.65\lambda$ and $\lambda_r \approx 0.35\lambda$.

Master Equations

Regular States (i, j) with $i, j < 3$

For states where neither player is at 40 yet:

$$\frac{d}{dt}p_{i,j}(t) = \lambda_s p_{i-1,j}(t) + \lambda_r p_{i,j-1}(t) - \lambda p_{i,j}(t),$$

where $p_{i-1,j} = 0$ if $i = 0$ and $p_{i,j-1} = 0$ if $j = 0$.

Examples:

$$\frac{d}{dt}p_{0,0}(t) = -\lambda p_{0,0}(t),$$

$$\frac{d}{dt}p_{1,0}(t) = \lambda_s p_{0,0}(t) - \lambda p_{1,0}(t),$$

$$\frac{d}{dt}p_{1,1}(t) = \lambda_s p_{0,1}(t) + \lambda_r p_{1,0}(t) - \lambda p_{1,1}(t).$$

Edge States: One Player at 40

For $(3, j)$ with $j < 3$ (server at 40):

$$\frac{d}{dt}p_{3,j}(t) = \lambda_s p_{2,j}(t) + \lambda_r p_{3,j-1}(t) - \lambda p_{3,j}(t).$$

Server scoring leads to Win-S (absorbed), so that flow leaves the system.

For $(i, 3)$ with $i < 3$ (receiver at 40):

$$\frac{d}{dt}p_{i,3}(t) = \lambda_s p_{i-1,3}(t) + \lambda_r p_{i,2}(t) - \lambda p_{i,3}(t).$$

Deuce and Advantage States

Deuce:

$$\frac{d}{dt}p_{\text{Deuce}}(t) = \lambda_s p_{2,3}(t) + \lambda_r p_{3,2}(t) + \lambda_r p_{\text{Ad-S}}(t) + \lambda_s p_{\text{Ad-R}}(t) - \lambda p_{\text{Deuce}}(t).$$

Flows into deuce come from:

$$(2,3) \xrightarrow{\lambda_s} \text{Deuce}, \quad (3,2) \xrightarrow{\lambda_r} \text{Deuce}, \quad \text{Ad-S} \xrightarrow{\lambda_r} \text{Deuce}, \quad \text{Ad-R} \xrightarrow{\lambda_s} \text{Deuce}.$$

Advantage Server:

$$\frac{d}{dt}p_{\text{Ad-S}}(t) = \lambda_s p_{\text{Deuce}}(t) - \lambda p_{\text{Ad-S}}(t).$$

Advantage Receiver:

$$\frac{d}{dt}p_{\text{Ad-R}}(t) = \lambda_r p_{\text{Deuce}}(t) - \lambda p_{\text{Ad-R}}(t).$$

Absorbing States

Server Wins:

$$\frac{d}{dt}p_{\text{Win-S}}(t) = \lambda_s p_{3,0}(t) + \lambda_s p_{3,1}(t) + \lambda_s p_{3,2}(t) + \lambda_s p_{\text{Ad-S}}(t).$$

Receiver Wins:

$$\frac{d}{dt}p_{\text{Win-R}}(t) = \lambda_r p_{0,3}(t) + \lambda_r p_{1,3}(t) + \lambda_r p_{2,3}(t) + \lambda_r p_{\text{Ad-R}}(t).$$

Full time-dependent solution

Setup and initial condition. We consider the continuous-time Markov model described above. The total event rate is

$$\lambda = \lambda_s + \lambda_r,$$

with λ_s the server-point rate and λ_r the receiver-point rate. We take the initial condition

$$p_{0,0}(0) = 1, \qquad p_{\text{anything else}}(0) = 0,$$

i.e. the game starts at 0–0 at time $t = 0$.

We denote by $p_{i,j}(t)$ the probability to be in the regular state (i, j) (server points i, receiver points j) with $i, j \in \{0, 1, 2, 3\}$ (note: $(3, 3)$ is the base deuce-count and will be related to the "Deuce" state below), by $p_{\text{Deuce}}(t)$ the probability mass in the Deuce state, by $p_{\text{Ad-S}}(t)$ and $p_{\text{Ad-R}}(t)$ the advantage states, and by $p_{\text{Win-S}}(t), p_{\text{Win-R}}(t)$ the absorbing win states. The master equations are linear ODEs. We now solve them explicitly.

1. Regular interior states (i, j) with $0 \leq i, j \leq 3$ (closed-form product form)

First we show that for all $i, j \in \{0, 1, 2, 3\}$ with the interpretation that these are the counts at time t (i.e. neither player has exceeded 3 yet at that time), the probabilities are the product of independent Poisson probabilities:

$$\boxed{p_{i,j}(t) \;=\; e^{-\lambda t}\,\frac{(\lambda_s t)^i}{i!}\,\frac{(\lambda_r t)^j}{j!}\,, \qquad 0 \leq i, j \leq 3.}$$

(As a remark: the right-hand side is exactly $P(N_s(t) = i, N_r(t) = j)$ where N_s, N_r are independent Poisson processes with rates λ_s, λ_r; when $i, j \leq 3$ this event implies the game has not yet been absorbed by someone reaching 4 earlier, so it equals the transient probability for the state (i, j).)

Proof (induction using integrating factor). The master equation for a generic regular state (i,j) with $i,j \geq 0$ (and with the convention $p_{-1,j} \equiv 0$, $p_{i,-1} \equiv 0$) is

$$\frac{d}{dt}p_{i,j}(t) = \lambda_s p_{i-1,j}(t) + \lambda_r p_{i,j-1}(t) - \lambda p_{i,j}(t).$$

Multiply by the integrating factor $e^{\lambda t}$ to get

$$\frac{d}{dt}\big(e^{\lambda t}p_{i,j}(t)\big) = \lambda_s e^{\lambda t}p_{i-1,j}(t) + \lambda_r e^{\lambda t}p_{i,j-1}(t).$$

We check base cases:

$$p_{0,0}(t): \quad \frac{d}{dt}p_{0,0} = -\lambda p_{0,0}, \qquad p_{0,0}(0) = 1$$

so $p_{0,0}(t) = e^{-\lambda t}$, consistent with the claimed formula.

Assume for all pairs with total index sum $i+j \leq n-1$ the formula holds. For any (i,j) with $i+j=n$ we integrate:

$$e^{\lambda t}p_{i,j}(t) = \int_0^t \big(\lambda_s e^{\lambda u}p_{i-1,j}(u) + \lambda_r e^{\lambda u}p_{i,j-1}(u)\big)\,du,$$

and by the induction hypothesis each integrand is

$$e^{\lambda u}p_{i-1,j}(u) = \frac{(\lambda_s u)^{\,i-1}}{(i-1)!}\frac{(\lambda_r u)^{\,j}}{j!}, \qquad e^{\lambda u}p_{i,j-1}(u) = \frac{(\lambda_s u)^{\,i}}{i!}\frac{(\lambda_r u)^{\,j-1}}{(j-1)!}.$$

So

$$e^{\lambda t}p_{i,j}(t) = \int_0^t \left(\lambda_s \frac{(\lambda_s u)^{\,i-1}}{(i-1)!}\frac{(\lambda_r u)^{\,j}}{j!} + \lambda_r \frac{(\lambda_s u)^{\,i}}{i!}\frac{(\lambda_r u)^{\,j-1}}{(j-1)!}\right) du.$$

Factor common terms and note the integrand equals the derivative (w.r.t. u) of

$$\frac{(\lambda_s u)^i}{i!}\frac{(\lambda_r u)^j}{j!},$$

because

$$\frac{d}{du}\left(\frac{(\lambda_s u)^i}{i!}\frac{(\lambda_r u)^j}{j!}\right) = \frac{i+j}{u}\frac{(\lambda_s u)^i}{i!}\frac{(\lambda_r u)^j}{j!}.$$

and algebra yields equality with the integrand above. So

$$e^{\lambda t}p_{i,j}(t) = \frac{(\lambda_s t)^i}{i!}\frac{(\lambda_r t)^j}{j!} - 0,$$

so

$$p_{i,j}(t) = e^{-\lambda t}\frac{(\lambda_s t)^i}{i!}\frac{(\lambda_r t)^j}{j!},$$

which completes the induction.

In particular the formulas give closed forms for all the regular states listed above:

$$p_{0,0}(t) = e^{-\lambda t}, \qquad p_{1,0}(t) = \lambda_s t e^{-\lambda t}, \qquad p_{0,1}(t) = \lambda_r t e^{-\lambda t},$$

$$p_{2,0}(t) = \frac{(\lambda_s t)^2}{2!} e^{-\lambda t}, \ \ldots, \ p_{3,2}(t) = \frac{(\lambda_s t)^3}{3!} \frac{(\lambda_r t)^2}{2!} e^{-\lambda t}, \text{ etc.}$$

2. Deuce and Advantage states — integral representations and closed-form approach

We now solve the ODEs for the Deuce / Advantage states. The master equations (as stated above) are

$$\frac{d}{dt} p_{\text{Deuce}}(t) = \lambda_s p_{2,3}(t) + \lambda_r p_{3,2}(t) + \lambda_r p_{\text{Ad-S}}(t) + \lambda_s p_{\text{Ad-R}}(t) - \lambda p_{\text{Deuce}}(t), \tag{1}$$

$$\frac{d}{dt} p_{\text{Ad-S}}(t) = \lambda_s p_{\text{Deuce}}(t) - \lambda p_{\text{Ad-S}}(t), \tag{2}$$

$$\frac{d}{dt} p_{\text{Ad-R}}(t) = \lambda_r p_{\text{Deuce}}(t) - \lambda p_{\text{Ad-R}}(t). \tag{3}$$

The inhomogeneous source terms $p_{2,3}(t), p_{3,2}(t)$ are known explicitly from the product-Poisson formulas above:

$$p_{2,3}(t) = e^{-\lambda t} \frac{(\lambda_s t)^2}{2!} \frac{(\lambda_r t)^3}{3!}, \qquad p_{3,2}(t) = e^{-\lambda t} \frac{(\lambda_s t)^3}{3!} \frac{(\lambda_r t)^2}{2!}.$$

Solve advantage equations via integrating factor. Equations (2), (3) are linear first-order ODEs with source $\lambda_s p_{\text{Deuce}}$ and $\lambda_r p_{\text{Deuce}}$. Use integrating factor $e^{\lambda t}$:

$$\frac{d}{dt}\left(e^{\lambda t} p_{\text{Ad-S}}(t)\right) = \lambda_s e^{\lambda t} p_{\text{Deuce}}(t).$$

Integrate from 0 to t (initial conditions $p_{\text{Ad-S}}(0) = p_{\text{Ad-R}}(0) = 0$):

$$e^{\lambda t} p_{\text{Ad-S}}(t) = \lambda_s \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du,$$

so

$$\boxed{p_{\text{Ad-S}}(t) = \lambda_s e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du}$$

and similarly

$$\boxed{p_{\text{Ad-R}}(t) = \lambda_r e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du}.$$

(These relations express the advantage-state masses as the convolutions of the Deuce mass with exponential kernels; they will be substituted into the Deuce equation.)

Single integral/Volterra equation for Deuce. Substitute the expressions for $p_{\text{Ad-S}}$ and $p_{\text{Ad-R}}$ into (1). Define the known forcing term

$s(t) := \lambda_s p_{2,3}(t) + \lambda_r p_{3,2}(t)$ (explicit known polynomial-times-$e^{-\lambda t}$ function).

Then (1) becomes

$$\frac{d}{dt} p_{\text{Deuce}}(t) + \lambda p_{\text{Deuce}}(t) = s(t) + \lambda_r p_{\text{Ad-S}}(t) + \lambda_s p_{\text{Ad-R}}(t).$$

Using the expressions for the advantage states,

$$\lambda_r p_{\text{Ad-S}}(t) + \lambda_s p_{\text{Ad-R}}(t) = (\lambda_r \lambda_s + \lambda_s \lambda_r)\, e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du = 2\lambda_s \lambda_r\, e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du.$$

So the Deuce equation becomes the Volterra integro-differential equation

$$\boxed{\frac{d}{dt} p_{\text{Deuce}}(t) + \lambda p_{\text{Deuce}}(t) = s(t) + 2\lambda_s \lambda_r\, e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du}.$$

Transform to a second-order linear ODE. Define

$$x(t) := e^{\lambda t} p_{\text{Deuce}}(t), \qquad y(t) := \int_0^t x(u)\, du = \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du.$$

Note $x(0) = 0,\ y(0) = 0$. Multiply the Deuce equation *by* $e^{\lambda t}$ to obtain

$$\frac{d}{dt} x(t) = e^{\lambda t} s(t) + 2\lambda_s \lambda_r\, y(t).$$

Differentiate once more:

$$\frac{d^2}{dt^2} x(t) = \frac{d}{dt}\big(e^{\lambda t} s(t)\big) + 2\lambda_s \lambda_r\, \frac{d}{dt} y(t).$$

But $\frac{d}{dt} y(t) = x(t)$. Therefore

$$\boxed{x''(t) - 2\lambda_s \lambda_r\, x(t) \;=\; \frac{d}{dt}\big(e^{\lambda t} s(t)\big)}.$$

This is a linear nonhomogeneous second-order ODE with constant coefficients. Define

$$\omega := \sqrt{2\lambda_s \lambda_r} \quad (\omega \geq 0).$$

Then

$$x''(t) - \omega^2 x(t) = f(t), \qquad f(t) := \frac{d}{dt}\big(e^{\lambda t}s(t)\big),$$

with initial conditions $x(0) = 0$, $x'(0) = e^{\lambda\cdot 0}s(0) + 2\lambda_s\lambda_r y(0) = 0$ (because $s(0) = 0$, $y(0) = 0$).

Solution by Green's function / convolution. The Green's function (causal) for the operator $L := \frac{d^2}{dt^2} - \omega^2$ with homogeneous initial conditions is

$$G(t) = \frac{1}{\omega}\sinh(\omega t) \quad (t \geq 0),$$

because $G'' - \omega^2 G = 0$ for $t > 0$ and the jump conditions match the delta forcing. So the particular solution is the convolution

$$x(t) = \int_0^t G(t-u)f(u)\,du = \int_0^t \frac{1}{\omega}\sinh\big(\omega(t-u)\big)\,f(u)\,du.$$

Recalling $p_{\text{Deuce}}(t) = e^{-\lambda t}x(t)$, we have the explicit representation

$$\boxed{p_{\text{Deuce}}(t) = e^{-\lambda t}\,\frac{1}{\omega}\int_0^t \sinh\big(\omega(t-u)\big)\,\frac{d}{du}\big(e^{\lambda u}s(u)\big)\,du}.$$

Integration-by-parts form (often more convenient). Integrate by parts in the integral above to remove the derivative on $f(u) = \frac{d}{du}(e^{\lambda u}s(u))$. Using

$$\int_0^t \sinh(\omega(t-u))f'(u)\,du = \Big[\sinh(\omega(t-u))f(u)\Big]_{u=0}^{u=t} + \omega\int_0^t \cosh(\omega(t-u))f(u)\,du,$$

and noting the boundary term simplifies (because $\sinh(0) = 0$ and $f(0) = 0$), one obtains an alternative representation

$$p_{\text{Deuce}}(t) = e^{-\lambda t}\int_0^t \cosh\big(\omega(t-u)\big)\,e^{\lambda u}s(u)\,du.$$

(One can check the algebra by carrying out the integration by parts carefully; both integral representations are equivalent and either can be used for explicit computation.)

Practical remark. The function $s(t) = \lambda_s p_{2,3}(t) + \lambda_r p_{3,2}(t)$ is explicit and is a polynomial in t multiplied by $e^{-\lambda t}$. Therefore $e^{\lambda u}s(u)$ is a polynomial in u and the integrals above reduce to finite combinations of exponentials and hyperbolic sines/cosines (i.e. closed-form elementary expressions). For brevity we keep the convolution representation which is compact and exact.

3. Express $p_{\text{Ad-S}}$, $p_{\text{Ad-R}}$ (re-stating) and absorbing flows

Once $p_{\text{Deuce}}(t)$ is known from the previous step, substitute back to get the advantage states (recall we already derived these integrals):

$$\boxed{p_{\text{Ad-S}}(t) = \lambda_s e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du, \qquad p_{\text{Ad-R}}(t) = \lambda_r e^{-\lambda t} \int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\, du}.$$

Absorbing-state ODEs and their integral solutions. The absorbing Win states satisfy

$$\frac{d}{dt} p_{\text{Win-S}}(t) = \lambda_s \big(p_{3,0}(t) + p_{3,1}(t) + p_{3,2}(t) + p_{\text{Ad-S}}(t)\big),$$
$$\frac{d}{dt} p_{\text{Win-R}}(t) = \lambda_r \big(p_{0,3}(t) + p_{1,3}(t) + p_{2,3}(t) + p_{\text{Ad-R}}(t)\big),$$

with zero initial conditions. Integrate in time from 0 to t to obtain

$$\boxed{p_{\text{Win-S}}(t) = \lambda_s \int_0^t \big(p_{3,0}(u) + p_{3,1}(u) + p_{3,2}(u) + p_{\text{Ad-S}}(u)\big)\, du}$$

and

$$\boxed{p_{\text{Win-R}}(t) = \lambda_r \int_0^t \big(p_{0,3}(u) + p_{1,3}(u) + p_{2,3}(u) + p_{\text{Ad-R}}(u)\big)\, du.}$$

Here the integrands are either explicit product-Poisson terms (for $p_{3,0}, p_{3,1}, p_{3,2}, p_{0,3}, p_{1,3}, p_{2,3}$) or the advantage-state convolution expressions above. So the absorbing probabilities are given by explicit time integrals of known functions; again closed-form elementary antiderivatives exist because the integrands are combinations of polynomials in u, exponentials $e^{-\lambda u}$ and the functions p_{Deuce} which themselves are convolution integrals with polynomial kernels. In consequence the absorptions $p_{\text{Win-S}}(t), p_{\text{Win-R}}(t)$ admit closed-form expressions in terms of exponentials, hyperbolic sines/cosines, and polynomials.

4. Consistency check and long-time limits

(i) Conservation of probability: for every $t \geq 0$ the total probability sums to 1:

$$\sum_{i=0}^{3} \sum_{j=0}^{3} p_{i,j}(t) \;+\; p_{\text{Deuce}}(t) + p_{\text{Ad-S}}(t) + p_{\text{Ad-R}}(t) \;+\; p_{\text{Win-S}}(t) + p_{\text{Win-R}}(t) \;=\; 1,$$

and this is guaranteed because our ODE system is that of an honest continuous-time Markov chain.

(ii) Long-time (eventual) probabilities: as $t \to \infty$ the transient masses vanish and all mass moves to the absorbing states; the limits give the eventual probabilities of server or receiver winning the game:

$$\lim_{t\to\infty} p_{\text{Win-S}}(t) \;=\; P(\text{server wins game}), \qquad \lim_{t\to\infty} p_{\text{Win-R}}(t) \;=\; P(\text{receiver wins game}).$$

One may recover the familiar discrete-time formulas (pre-deuce + deuce contributions) by carrying the integrals to ∞ and evaluating; the decomposition and evaluation are consistent with the combinatorial formulas shown earlier (e.g. $P_{\text{pre-deuce}} = p^4(1+4q+10q^2)$ and $P_{\text{deuce}\to\text{win}} = p^2/(p^2+q^2)$ when writing $p = \lambda_s/\lambda$, $q = \lambda_r/\lambda$).

5. Summary of explicit expressions (compact list)

- **Regular states (i,j), $0 \le i,j \le 3$:**

$$p_{i,j}(t) = e^{-\lambda t}\frac{(\lambda_s t)^i}{i!}\frac{(\lambda_r t)^j}{j!}.$$

- **Advantage states:**

$$p_{\text{Ad-S}}(t) = \lambda_s e^{-\lambda t}\int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\,du, \qquad p_{\text{Ad-R}}(t) = \lambda_r e^{-\lambda t}\int_0^t e^{\lambda u} p_{\text{Deuce}}(u)\,du.$$

- **Deuce:** with $\omega = \sqrt{2\lambda_s\lambda_r}$ and $s(t) = \lambda_s p_{2,3}(t) + \lambda_r p_{3,2}(t)$,

$$p_{\text{Deuce}}(t) = e^{-\lambda t}\,\frac{1}{\omega}\int_0^t \sinh\big(\omega(t-u)\big)\,\frac{d}{du}\big(e^{\lambda u}s(u)\big)\,du,$$

  or equivalently

$$p_{\text{Deuce}}(t) = e^{-\lambda t}\int_0^t \cosh\big(\omega(t-u)\big)\,e^{\lambda u}s(u)\,du.$$

  (Either form gives a closed-form expression since $e^{\lambda u}s(u)$ is a polynomial in u.)

- **Absorbing states:**

$$p_{\text{Win-S}}(t) = \lambda_s\int_0^t \big(p_{3,0}(u) + p_{3,1}(u) + p_{3,2}(u) + p_{\text{Ad-S}}(u)\big)\,du,$$

$$p_{\text{Win-R}}(t) = \lambda_r\int_0^t \big(p_{0,3}(u) + p_{1,3}(u) + p_{2,3}(u) + p_{\text{Ad-R}}(u)\big)\,du.$$

Summary of the Complete System

There are 20 states in total:

- 16 regular point states: (i, j) for $i, j \in \{0, 1, 2, 3\}$, including the deuce state with $(3, 3)$,
- 2 advantage states,
- 2 absorbing states.

The system forms a continuous-time Markov chain that can be written compactly as:

$$\frac{d\mathbf{p}}{dt} = A\mathbf{p},$$

where A is the rate matrix encoding all transitions.